MFCS Design Doc

This document defines the primary pages, layouts, controls, role-specific behavior, and user flows for the MCFS Learning Platform. The platform has three authenticated roles:

- **Student** — views assigned lessons, attempts practice activities, and sees only their own progress.
- **Teacher** — creates/manages lessons and views progress for students in classes they teach.
- **Admin** — manages users/classes and can view or modify content and progress across the entire school.

There is currently no parent account. Student-facing progress and assignment information should use enough English context that non-Chinese-speaking parents can understand what the student is expected to do.

Traditional Chinese should be the default Chinese display form. Mandarin and Cantonese pronunciation must be clearly distinguished where relevant. Mandarin pronunciation should use Pinyin, while Cantonese pronunciation should use Jyutping.

1. Global Navigation & Shared UI

Logged-Out Header

- Displays for Homepage, Log In Page, Contact Support Page
- Left-hand side: MCFS logo/name
- Right-hand side: Home, Log In, Contact Support
- On mobile, these can collapse into a hamburger menu.

Logged-In Header

- Displays for all other pages
- Left-hand side: MCFS logo/name, Dashboard
- Right-hand side: User's Name, Account, Log Out
- On mobile, these can collapse into a hamburger menu.

Lessons, Students, and Teachers (for Admins) will be viewable on the Dashboard page (see below)

Shared Design Principles

Student UI should emphasize large clickables, simple wording, clear and visual progress, and mobile-friendly controls. Teacher/Admin UI can be denser and more management-oriented.

2. Homepage

Access: Everyone, including users who cannot log in
Purpose: Simple public landing page before authentication.

Layout:

- "MCFS Learning Platform", title
- Short written explanation (users can review weekly lessons, practice from home, track learning progress, etc.)
- "Log In" Button

Primary Navigation Actions:

- Log In button → Log-In Page

3. Contact Support Page

Access: Everyone, including users who cannot log in

Purpose: Allow students, parents, teachers, or others to contact MCFS administrators for technical support.

Layout:

- "Contact Support" Title
- Support Form with fields:
    - Name
    - Email
    - Role (Student/Parent, Teacher)
    - Subject
    - Description of issue
- "Submit Request" button: sends form results to all administrator emails, leading to one of the following states:
    - Success: "Your support request has been submitted" displayed on screen
    - Error: "There was an error submitting your request. Please contact <email TBD>" displayed on screen

Primary Navigation Actions:

- N/A

4. Log-In Page

Access: Logged-out users
Purpose: Authenticate Students, Teachers, and Admins.

Layout:

- Centered Login Card containing:
    - MCFS logo
    - Username or email section

    - Password section
    - Show/hide password control
- Log In button: authenticates user and allows system to determine user role, leading to one of two states:
    - Authentication Success: (appropriate) Dashboard Page
    - Authentication Failure: "Incorrect Username or Password" displayed
- Forgot Password button: Activates Password Recovery Pop-up
- Contact Support button: Navigates to Contact Support Page

Primary Navigation Actions:

- Log In button → (appropriate) Dashboard Page
- Forgot Password button → Password Recovery Pop-up
- Contact Support button → Contact Support Page

4.1. Password Recovery Pop-up

Layout:

- Fields:
    - Username OR email
- Send Reset Instructions button: Behavior depends on:
    - Account Found: Sends the user a code to the account's associated email, and opens a new field for code entry. Entering the code leads to the Reset Password Pop-up
    - Account Not Found: "Username or email do not match with an account" displayed
- Back to Log In button: Navigates to Log-In Page

Primary Navigation Actions:

- Send Reset Instructions button → Reset Password Pop-up

- Back to Log In button → Log-In Page

4.2. Reset Password Pop-up

Layout:

- Fields:
    - New Password
    - Confirm Password
    - Reset Password button: Behavior depends on:
        - Fields Match: Changes the user's current password to the entered one, and sends them to the Log-In Page with "Password successfully reset" displayed
        - Fields Don't Match: "Fields do not match" displayed

Primary Navigation Actions:

- Reset Password Button → Log-In Page

5. Account / Profile Page

Access: All authenticated users

Purpose: Provide basic account information for a user

Layouts:

- Student:
    - Displays Full Name, Username, Email, Class, Teacher
    - Cannot be edited
- Teacher:
    - Displays Full Name, Username, Email, Class(es)
    - Username is the only field that can be changed
- Admins:
    - Displays Full Name, Username, Email

    - All info can be changed
- (Shared) Buttons:
    - Change Password: Opens Forgot Password Pop-up
    - Log Out: Logs user out and redirects to Homepage

Primary Navigation Actions:

- Change Password Button → Reset Password Pop-up
- Log Out Button → Homepage

6. Student Dashboard Page

Access: Student
Purpose: Display current week's lessons, previous weeks' lessons, and current progress

Layout

- Welcome Section
    - "Welcome, <FirstName>!"
    - Class name, Teacher
- My Progress Section
    - Includes widgets for lessons completed, lessons remaining, recent scores, etc.
- Lessons Section
    - Shows current week's lesson first, then previous weeks
    - Each lesson allows you to click on the name to open the Lesson Card for that lesson

Primary Navigation Actions:

- Lessons Section Buttons → (that lesson's specific) Lesson Card Pop-up

6.1. Student Lesson Card Pop-up

Layout:

- Body, including:
    - Lesson Title
    - Week #
    - Short description
    - Assignment status (Not Started, In Progress Completed)
    - Latest score (or N/A)
- Open Lesson: leads to the lesson's Lesson Attempt Page
- Results button: Opens the lesson's Lesson Results Page

Primary Navigation Actions:

- Open Lesson Button → Lesson Attempt Page
- Results Button: Lesson Results Page

7. Teacher Dashboard Page

Access: Teacher

Purpose: Display an overview of the teacher's current lessons and students' progress

Layout

- Overview Section
    - Shows a quick view summarizing activity the teacher's most recently created lesson, example:

        Week 7 — Food Vocabulary
        18 students assigned
        13 completed
        5 not completed

- Two Switchable Tabs:
    - Lessons Section
        - Create Lesson button at the top
        - Displays only lessons the teacher has created in chronological order (new to old)
        - Each lesson displays its status (Draft/Published)
            - Draft lessons are only visible to the Teacher who created them and Admins
        - Lessons are clickable, leading to the lesson's card pop-up
    - Students Section
        - Shows a table of all students where each column is sortable in ascending/descending order, including:
            - Student Name
            - Class
            - % of lessons completed
            - Latest lesson completed
        - Students are clickable, leading to the student's progress pop-up
        - Includes filtering, sorting, etc.

Primary Navigation Actions:

- Create Lesson button: Lesson Creator Page
- Lessons Section button(s): Teacher Lesson Card Pop-up
- Student Section button(s): Teacher Analytics Pop-up

7.1. Teacher/Admin Lesson Card Pop-up

Layout:

- Body:

    - Lesson Title
    - Status (Draft/Published)
    - Week #
    - Assigned Classes
    - Completion count
- Edit button: leads to Lesson Creator Page for that lesson
- Preview button: leads to Lesson Preview Page for that lesson
- Duplicate button: leads to a new Lesson Creator Page for a new lesson with the same field data

Primary Navigation Actions:

- Edit button → Lesson Creator Page
- Preview button → Lesson Preview Page
- Duplicate button → Lesson Creator Page

7.2. Teacher/Admin Analytics Pop-up

Layout:

- Student's Name
- Student's Class
- % of Lessons Completed
- All assigned lessons, displaying:
    - Lesson Name
    - Status
    - Latest score
    - Best score
    - # of attempts

8. Admin Dashboard Page

Access: Admin

Purpose: Admin uses the same general visual system as Teacher but with school-wide scope.

Layout:

- Overview Section
    - Shows a quick view summarizing site-wide activity, including total students, teachers, classes, and lessons
- Four Switchable Tabs:
    - Lessons Section
        - Same as Teacher Dashboard Page, but lessons also signify which teacher created them next to the name of the lesson
    - Students Section
        - Same as Teacher Dashboard Page, but there is an "Edit User" button next to each student in the table, opening the Admin Edit User Pop-up
        - Includes pagination for fetch requests due to the large amount of students
    - Teachers Section
        - Similar to Student Section, with columns for:
            - Name
            - Username/Email
            - Assigned Class(es)
        - Includes an "Edit User" button next to each teacher in the table, opening the Admin Edit User Pop-up
    - Classes Section
        - Create Class button: opens Admin Class Pop-up
        - Edit Class button: opens Admin Class Pop-up linked to the class' info, allowing editing of the class info

        - Delete Class button: Allows deletion of a class after a confirmation
        - A table with a list of all class names vertically

Primary Navigation Actions:

- Create Lesson button: Lesson Creator Page
- Lessons Section button(s): Teacher/Admin Lesson Card Pop-up
- Edit User button: Admin Edit User Pop-up
- Student Section button(s): Teacher/Admin Analytics Pop-up
- Create Class button: Admin Edit Class Pop-up
- Edit Class button: Admin Edit Class Pop-up
- Delete Class button: Deletion Confirmation Pop-up

8.1. Admin Edit User Pop-up

Layout:

- Common Fields:
    - Display Name
    - Username
    - Email
    - Account Status (Active/Inactive)
    - Password
- Student-specific Fields:
    - Class
- Teacher-specific Fields:
    - Assigned Class(es)
- Save Changes button: Saves updated account information
- Delete Account button: Opens Deletion Confirmation Pop-up
- Cancel button: Closes pop-up without saving

Primary Navigation Actions:

- Save Changes button → Admin Dashboard Page
- Delete Account button → Deletion Confirmation Pop-up
- Cancel button → Admin Dashboard Page

8.2. Admin Edit Class Pop-up

Layout:

- Fields:
    - Class Name
    - Assigned Teacher(s)
    - Assigned Students
- Save Changes button: Saves the class information
- Cancel button: Closes pop-up without saving

Primary Navigation Actions:

- Save Changes button → Admin Dashboard Page
- Cancel button → Admin Dashboard Page

9. Lesson Attempt Page

Access: Student

Purpose: Display lesson information and previous attempts before starting or retrying a lesson.

Layout:

- Lesson Information:
    - Lesson Title
    - Week #
    - Short description/instructions
- Vocabulary Section:

    - Students can review the vocab in the lesson before beginning the questions, each vocab displays:
        - Traditional Chinese
        - Simplified Chinese, if included
        - Meaning(s)
        - Context Notes, if included
        - Mandarin pronunciation (Pinyin), if applicable
        - Cantonese pronunciation (Jyutping), if applicable
        - Example sentence, if included
- Attempt Status:
    - Not Yet Attempted, OR:
        - Latest Score
        - Best Score
        - Previous Attempts
- Start Lesson/Continue Lesson button: Begins the lesson
- Results button: Opens Lesson Results Page, only clickable if lesson has been attempted

Primary Navigation Actions:

- Start Lesson/Continue Lesson button → Lesson Question Page
- Results button → Lesson Results Page

10. Lesson Question Page

Access: Student

Purpose: Display one lesson question at a time.

Layout:

- Top Section:
    - Lesson Title
    - "Question # of #"

    - Progress Bar
    - Exit Lesson button: Exits back to the Lesson Attempt Page, saving progress up until that point
- Question Section changes depending on question type:
    - Character Recognition / Understanding:
        - Chinese character/word
        - Question/prompt
        - Multiple answer choices
    - Pronunciation Practice:
        - Chinese character/word
        - Pronunciation guide, clearly labeled as Mandarin (Pinyin) or Cantonese (Jyutping)
        - Play Audio button: Plays the correct pronunciation with highlighted characters
        - Record button: Allows the student to record their voice
        - Playback button: Plays-back the student's recording
    - Character Writing Practice:
        - Chinese character/word
        - Pronunciation guide, clearly labeled as Mandarin (Pinyin) or Cantonese (Jyutping)
        - Meaning/context
        - Stroke/tracing guide
        - HTML5 Canvas writing area
        - Clear button: Clear the canvas
- Shared Controls:
    - Submit button: Submits the current response and displays Question Feedback Pop-up

Primary Navigation Actions:

- Submit button → Question Feedback Pop-up
- Exit Lesson button → Lesson Attempt Page

10.1. Question Feedback Pop-up

Layout:

- Correct/Incorrect indicator (checkmark vs x-mark)
- Correct answer
- Correct answer's explanation
- Next Question button: Proceeds to the following Lesson Question Page
- Finish Lesson button: replaces Next Question on the final question, leading to the Lesson Results Page

Primary Navigation Actions:

- Next Question button → Lesson Question Page
- Finish Lesson button → Lesson Results Page

11. Lesson Results Page

Access: Student
Purpose: Display the student's results after completing a lesson attempt.

Layout:

- "Lesson Complete!" message
- Results Summary:
    - Score % and # of Questions Correct/Total Questions
    - Completion Date
- Results In-Depth:
    - Each question stacked vertically on top of one another

    - Shows a similar view to the Question Feedback Pop-up, explaining the correct answer
- Try Again button: sends user to first question page of that lesson
- Back to Dashboard button: sends user back to their dashboard

Primary Navigation Actions:

- Try Again button → Lesson Question Page
- Back to Dashboard button → Student Dashboard Page

12. Lesson Creator Page

Access: Teacher/Admin

Purpose: Allow Teachers and Admins to create or modify lessons.

Layout:

- Top Section:
    - Preview button: moves to the preview page for that lesson
    - Save Draft button: saves the lesson as a draft, bringing the user back to the dashboard without publishing the lesson
        - Has auto-save functionality, with a short textbox that reads that the data is being automatically saved
    - Publish button: publishes the lesson to the assigned class, and brings the user back to their dashboard
    - Delete Lesson button: opens a pop-up to delete the lesson
- Lesson Information Section:
    - Displays all information about the lesson, with fields:
        - Lesson Title
        - Status (Draft/Published)
        - Assigned Class(es)

                - Only shows classes under a teacher. Shows all classes for admins
            - Week #
            - Student Instructions
- Vocabulary Section:
    - Displays all vocabulary for the lesson in the order they were added
        - Each vocabulary item has a delete button, which opens the deletion confirmation pop-up for that question
    - Allows teachers to add more vocabulary, with the following fields:
        - Traditional Chinese
        - Optional Simplified Chinese
        - Meaning(s)
        - Optional Context Notes
        - Optional Mandarin Pronunciation (Pinyin)
        - Optional Cantonese Pronunciation (Jyutping)
        - Optional Example Sentence(s)
    - Add Vocabulary Item button: Adds the current vocabulary to the study-set for that lesson
- Questions Section:
    - Displays all questions in lesson order, with reordering controls
        - Each question item has a delete button, which opens the Deletion Confirmation Pop-up
    - Allows teachers to add more questions to the question set, with the following fields:
        - Vocabulary word (selectable menu containing all vocab words in the lesson)

        - Question Type:
            - Character Recognition / Understanding
            - Pronunciation Practice
            - Writing Practice
    - Add Question button: Adds the current question to the lesson

Primary Navigation Actions:

- Preview button: Lesson Preview Page
- Save Draft button: Teacher/Admin Dashboard Page
- Publish button: Teacher/Admin Dashboard Page
- Delete Lesson button: Deletion Confirmation Pop-up
- Delete button(s): Deletion Confirmation Pop-up

12.1. Deletion Confirmation Pop-up

Layout:

- Confirmation message "Are you sure you want to delete '<item being deleted>'? This action cannot be undone."
- Cancel button: closes the pop-up
- Confirm button: deletes the current item from data, and takes the user back to their appropriate dashboard

Primary Navigation Actions:

- Confirm button → Teacher/Admin Dashboard

13. Lesson Preview

Access: Teacher/Admin

Purpose: Allow Teachers/Admins to experience a lesson as a Student before publishing it.

Layout:

- Uses the same layout as:
    - Lesson Attempt Page
    - Lesson Question Page
    - Lesson Results Page
- Persistent message:
    - "Preview Mode — Student progress will not be recorded"
- Exit Preview button: Takes the user back to that lesson's creator page

Primary Navigation Actions:

- Exit Preview button → Lesson Creator Page

14. Primary User Flows

Student — Complete Lesson

Homepage
→ Log In Page
→ Student Dashboard Page
→ Select Lesson
→ Lesson Attempt Page
→ Review Vocabulary
→ Start Lesson
→ Lesson Question Page
→ Submit Answer
→ Question Feedback
→ Next Question
→ ...
→ Final Question

→ Lesson Results Page
→ Student Dashboard Page / Try Again

Student — Continue Incomplete Lesson
Student Dashboard Page
→ Select In-Progress Lesson
→ Lesson Attempt Page
→ Continue
→ Next Unanswered Lesson Question
→ Complete Remaining Questions
→ Lesson Results Page

Teacher — Create and Publish Lesson
Log In Page
→ Teacher Dashboard Page
→ Create Lesson
→ Lesson Creator Page
→ Enter Lesson Information
→ Add Vocabulary
→ Add/Reorder Questions
→ Save Draft
→ Preview Lesson
→ Publish
→ Teacher Dashboard Page

Teacher — Continue Editing Draft
Teacher Dashboard Page
→ Lessons Section
→ Select Draft Lesson

→ Teacher/Admin Lesson Card Pop-up
→ Edit
→ Lesson Creator Page
→ Make Changes
→ Save Draft OR Publish

Teacher — View Student Progress
Teacher Dashboard Page
→ Students Section
→ Select Student
→ Teacher/Admin Analytics Pop-up
→ Review Lesson Progress
→ Close

Teacher — Modify Published Lesson
Teacher Dashboard Page
→ Lessons Section
→ Select Published Lesson
→ Teacher/Admin Lesson Card Pop-up
→ Edit
→ Lesson Creator Page
→ Make Changes
→ Preview
→ Update

Admin — Manage Student
Admin Dashboard Page
→ Students Section
→ Select Student / Edit User

→ Admin Edit User Pop-up
→ Save Changes

Admin — Manage Teacher
Admin Dashboard Page
→ Teachers Section
→ Select Teacher / Edit User
→ Admin Edit User Pop-up
→ Save Changes

Admin — Manage Lesson
Admin Dashboard Page
→ Lessons Section
→ Select Draft or Published Lesson
→ Teacher/Admin Lesson Card Pop-up
→ Edit
→ Lesson Creator Page
→ Modify
→ Preview
→ Save Draft / Publish / Update

Admin — Manage Class
Admin Dashboard Page
→ Classes Section
→ Create/Edit Class
→ Admin Edit Class Pop-up
→ Save Changes

Role Permission Summary

Capability	Student	Teacher	Admin
View own dashboard	Yes	Yes	Yes
View assigned published lessons	Yes	–	–
View lesson vocabulary	Assigned lessons	Own lessons	All lessons
Attempt lessons	Assigned lessons	Preview only	Preview only
Continue incomplete attempts	Yes	–	–
View students and student progress	Own info	All info in own classes	All students
Create/Draft Lessons	No	Yes	Yes
View Draft lessons	No	Own drafts	All drafts
Publish/Edit/Delete lessons	No	Own lessons	All lessons
Edit student and teacher accounts	No	No	Yes
View teachers and manage classes	No	No	Yes